Alberto Hernández Espinosa
Mexico City, Mexico • +52 552 383 4438 • albertohernandezespinosa@gmail.com
https://albertoHdzE.github.io/website/ • https://linkedin.com/in/alberto-hernandeze •
https://github.com/albertoHdzE

1 Professional Summary

- Applied AI researcher and engineer with over 10 years of experience across academic research and production systems, building Python-based ML and agentic workflows for analysis, retrieval, and decision support.
- Experience designing multi-agent systems, local LLM workflows, RAG/GraphRAG pipelines, knowledge-graph workflows, and observability layers for evaluation, orchestration, and tool use.
- Work includes document understanding, extraction, layout analysis, multimodal evaluation and local fine-tuning workflows, quantitative trading systems, and benchmarking of retrieval pipelines under fixed evaluation workflows.
- Published with Oxford University and Oxford Immune Algorithmics on integrated information and algorithmic complexity; currently AI Engineer Lead at K-Square Group.

2 Skills & Expertise

- Agentic / RAG Systems: LangGraph, Ollama, ML Studio, RAG, GraphRAG, knowledge-graph workflows, Agents, MCP, Multi-Agent Orchestration, Local LLM Orchestration, Langfuse. Relevant projects: HiFi, GraphRAG, CouncilAI, ContextForge, Noesis, KS-Onboarding, AG-UI.
- Evaluation / Observability: telemetry, monitoring, observability, fixed-workflow benchmarking, retrieval comparison, multimodal evaluation, local fine-tuning workflows. Relevant projects: HiFi, CouncilAI, ContextForge.
- Python / ML Stack: Python, TensorFlow, PyTorch, Scikit-learn, Keras, SQL/NoSQL, ETL, Pandas, NumPy, Matplotlib. Relevant projects: CouncilAI, Noesis, Stock Prices Prediction, MOSAICA.
- Cloud / APIs: AWS, GCP, Docker, FastAPI, React. Relevant projects: HiFi, ContextForge, Noesis.
- Applied Areas: document understanding, extraction, layout analysis, document and meeting analysis with OCR-enabled workflows, quantitative trading, reinforcement learning, complexity science, algorithmic information theory. Relevant projects: KS-Onboarding, HiFi, Noesis, AG-UI, Stock Prices Prediction, CausalBool, MOSAICA.
- Programming Languages: Python, R, C++, SQL, Mathematica.

3 Professional Experience

3.1 AI Engineer Lead, K-Square Group (Remote)

Nov 2024–Present

- Conduct and lead teams for implementation of AI-driven projects, enhancing team collaboration and project execution.
- Spearhead research and development of AI/ML applications tailored to specific industry challenges, focusing on innovation and efficiency.
- Design and oversee the integration of advanced AI technologies, including agentic systems and complex data processing pipelines.

- Develop and implement strategic frameworks for rapid prototyping and proof-of-concept development, aligning with organizational goals.
- Foster a culture of experimentation and continuous improvement, mentoring teams in leveraging AI tools effectively.

3.2 Chief AI Scientist, MilkStraw AI, San Francisco, CA (Remote)

Oct 2022–Sep 2024

- Led research of AI-driven cloud resource optimization models on AWS, achieving up to 53% cost savings for clients.
- Designed and deployed ML pipelines using Python, TensorFlow, and PyTorch.
- Collaborated remotely with global teams to deliver scalable AI solutions.

3.3 Research Director, Laxford Capital (Remote)

Jun 2019–Jul 2022

- Directed AI trading initiatives, developing reinforcement learning agents for financial markets.
- Oversaw model design and implementation in Python and PyTorch.
- Enhanced trading performance through causal AI approaches.

3.4 Research Collaborator, Oxford Immune Algorithmics & Oxford University, Oxford, UK (Remote with Visits)

2015–Nov 2024

- Contributed to publications on integrated information theory and algorithmic complexity.
- Implemented models in Mathematica and Python.
- Co-authored papers with Oxford researchers, advancing cognitive science applications.

4 Education

4.1 PhD in Philosophy of Science, UNAM, Mexico City, Mexico

2015–2019 (Specialization: Artificial Intelligence and Cognitive science)

- Research focused on integrated information theory (Phi).
- CONACyT Scholarship.
- Education in association with University of Oxford, Computer Science Faculty.

4.2 Master in Computer Sciences, UNAM, Mexico City, Mexico

2012–2014 (Specialization: Artificial Intelligence)

- Thesis on computational creativity models.
- CONACyT Scholarship.

5 Selected Projects

- GraphRAG From Scratch
  - Scientific baseline for RAG, RAU, and GraphRAG that separates retrieval, graph construction, synthesis, and knowledge-graph workflows for evaluation.
  - Code: https://github.com/albertoHdzE/GraphRAG-scratch
- HiFi
  - Local multi-agent trading system with document understanding, multimodal evaluation, and local fine-tuning of LLM families through ML Studio, using MCP, ContextForge, Langfuse, and telemetry for deterministic strategy creation.
  - Code: https://github.com/albertoHdzE/HiFi
- CouncilAI V2 (AI POCs Feature Spanner)
  - Benchmark harness and dashboard for comparing vector, graph, knowledge-graph, long-context, and managed-search retrieval pipelines under a fixed agentic evaluation workflow.
  - Code: https://github.com/albertoHdzE/ai-pocs-feature-spanner
- ContextForge (k-mcp-farm)
  - Open-source registry and proxy that federates MCP, A2A, and REST/gRPC APIs with centralized governance, discovery, and observability.
  - Code: https://github.com/albertoHdzE/k-mcp-farm
- AG-UI
  - Agent-user interaction layer for observable human-in-the-loop interfaces supporting document-analysis workflows in agentic applications.
  - Code: https://github.com/albertoHdzE/agui
- CausalBool: Deterministic Causal Integration
  - Physics-inspired algorithmic information framework for Boolean causal networks with deterministic causal reconstruction and scalable information signatures.
  - Code: https://github.com/albertoHdzE/CausalBool
- Noesis: Agentic Agnostic Platform
  - Agentic platform for building AI/ML models with LangGraph, Ollama, FastAPI, and React, including extraction-oriented workflows applied to fraud detection.
  - Code: https://github.com/KSProjectX/test-xx (restricted access).
- KS-Onboarding: Agentic Platform for Project Control and Onboarding
  - Agentic system for project and document analysis using OCR, extraction, layout-aware processing, voice recognition, and advanced RAG over local/external information.
  - Code: https://github.com/KSProjectX/KS-onboarding (restricted access).
- Agents for Automation of Running Tests
  - Agentic platform that turns user stories or code into pytest suites and coverage reports.
  - Code: https://github.com/Sucharita8/KS-qe-platform (restricted access).
- Stock Prices Prediction

- Deep reinforcement learning and ML workflows for stock price prediction and autonomous trading decisions.
- GitHub: https://github.com/albertoHdzE/finance_1, Detailed explanation: https://sites.google.com/view/complexai/FINANCE.

- MOSAICA: DNA Complexity Analysis
  - Machine learning and algorithmic complexity analysis of DNA sequences, linking genomic information content to DNA curvature.
  - GitHub: https://github.com/albertoHdzE/MOSAICA.

6 Publications

- 2026: "SuperARC: An Agnostic Test for Narrow, General, and Super Intelligence Based On the Principles of Recursive Compression and Algorithmic Probability." Nature. https://www.nature.com/articles/s41467-026-73289-5
- 2025: "Neurodivergent Influenceability as a Contingent Solution to the AI Alignment Problem." PNAS. https://academic.oup.com/pnasnexus/article/5/4/pgag076/8651394
- 2021: "Estimations of Integrated Information Based on Algorithmic Complexity and Dynamic Querying." World Scientific. https://www.worldscientific.com/doi/10.1142/9789811235726_0005
- 2017: "Is There Any Real Substance to the Claims for a 'New Computationalism'?" Springer. https://link.springer.com/chapter/10.1007/978-3-319-58741-7_2
- 2013: "Does the Principle of Computational Equivalence Overcome the Objections Against Computationalism?" Springer. https://link.springer.com/chapter/10.1007/978-3-642-37225-4_14

7 Certifications

- Mathematics for Machine Learning
- AI for Finance
- Google Cloud Specialization
- Reinforcement Learning
- Complexity Science

Full list of 47 certifications available here: https://www.linkedin.com/in/alberto-hernandeze/details/certifications/

8 Languages

- Spanish (mother language)
- English (advanced proficiency)
- German (Basic)
- French (Basic)
- Japanese (Basic)

Certifications available here: https://www.linkedin.com/in/alberto-hernandeze/details/languages/